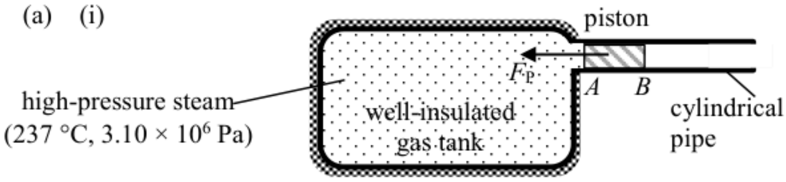
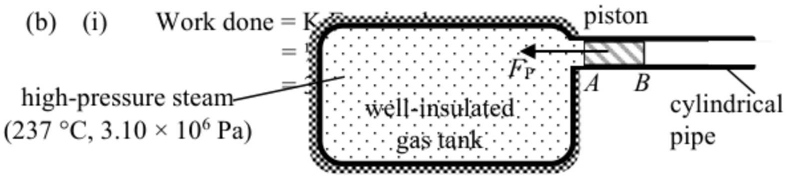
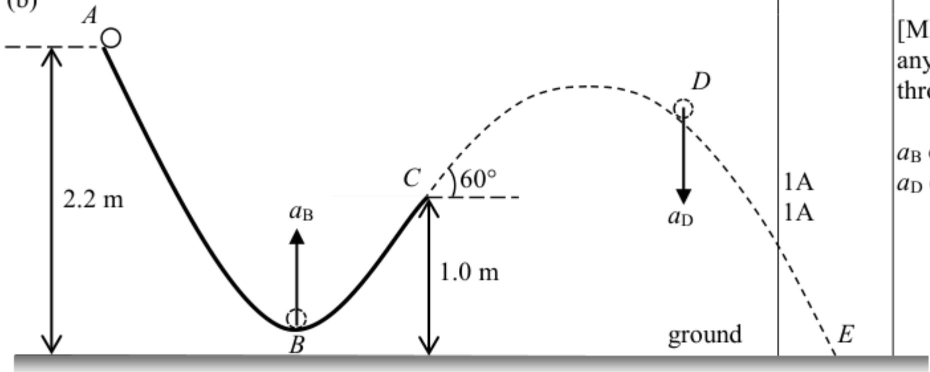
1. (a) $5 \times 0.02 \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$	1M	Accept 85.0 °C to 85.4 °C
$T = 85.347368\ °C \approx 85.3\ °C$	1A	[Marking Remark (MR): temperature difference must be correct &
	2	accept missing '5' – 1 M]
(b) (i) To compensate for / balance the heat loss (of the container with soup) to the surroundings.	1A	[MR: Key words - heat loss to the surroundings (from container or
	1	through evaporation)]
(ii) $P \times 10 \times 60 = 2000 \times 9 + 16 \times 4200 \times 9$	1M+1M	Assume :
$P = 1038\ W \approx 1040\ W$	1A	Power of the heater = Rate of heat loss (of the container with soup) to the surroundings
	3	[MR: $P \times t$ ✓ + 1 term ✓ - 1M $P \times t$ ✓ + 2 term ✓ - 2M $P \times t$: accept $P \times 10$ ΔT: accept $(96 - 9)$]
(iii) Smaller than 9 °C.	1A	
As the temperature of (the container with) soup drops, the rate of heat loss to the surroundings becomes lower due to a smaller temperature difference (w.r.t. the surroundings).	1A	[2 A marks are independent Key concept: smaller temperature difference]
	2	

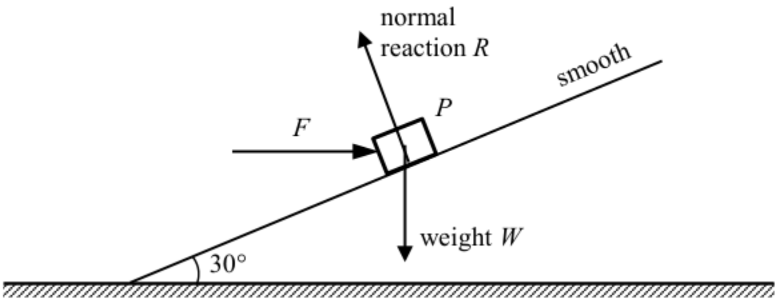
2. (a) (i) 	1A	F_P correctly marked F_P should be in contact with the piston. Accept: no label of the force
	1	
(ii) $F_P = (3.10\times10^6 - 1.0\times10^5)\times0.67$	1M	Accept: $F_P = (3.10\times10^6)\times0.67$
$= 2010000\text{ N} = 2.01\times10^6\text{ N}$	1A	Accept 2.00×10^6 N to 2.01×10^6 N
	2	
(iii) $pV = nRT \Rightarrow V = \dfrac{nRT}{p}$ $V = \dfrac{(570/0.018)(8.31)(237+273)}{3.10\times10^6}$	1M+1M	1st 1M find n (570/0.018) 2nd 1M $pV=nRT$, accept T=237
$= 43.292419\text{ m}^3 \approx 43.3\text{ m}^3$	1A	Accept 43.0 m³ to 43.3 m³
	3	
(b) (i) Work done = K = = 	1M 1A	Accept 37.9 MJ to 38.0 MJ Accept: $a = 54/1.5 = 36$ (m s^{-2}) $F = (2.6\times10^4)\times36$ $= 936000$ (N) $s = 54^2/(2\times36) = 40.5$ (m) [or $v = at/2 = 36(1.5)/2$ $= 27$ (m s^{-1})] $W = Fs = 936000\times340.5$ [or $W = Fvt = 936000\times27\times1.5$] $= 37908000$ (J) [1M+1A] If $s = vt$ [0M 0A]
	2	
(ii) Average acceleration $a = \dfrac{v-u}{t} = \dfrac{54-0}{1.5}$	1M	
$= 36\text{ m s}^{-2}$	1A	
	2	
(iii) Acceleration is decreasing (i.e. maximum at first). (According to kinetic theory,) once the steam expands, i.e. volume increases, its pressure decreases, thus the (pressure) force acting on the piston at A decreases, hence a smaller acceleration.	1A 1A 1A	1st 1A for acceleration decreasing 2nd 1A for volume increases/ length increases ⇒ pressure decreases Alternative method: The acceleration of the jet fighter is decreasing, 1A As the air resistance (f) acts on the jet fighter increases with the speed. 1A The net force ($ma = F - f$) is decreasing. 1A
	3	

3. (a)	With superconductor / extremely low or nearly zero resistance, a much larger current flows in the coils producing a stronger magnetic field (with less heat loss due to the current flow).	1A 1A		[MR: low R + large I – 2A marks large I only – 1 A mark low R only – 0 A mark]
			2	
(b)	P and Q : N (north pole) Repulsive force between like poles	1A 1A		
			2	
(c) (i)	As the train is not in contact with the rail, <u>there is no friction / interaction between the train and the rail</u>, thus smoother due to less vibration.	1A		
(ii)	There is no friction / interaction between the train and the rail, and the propulsion of the train only needs to work against air resistance, therefore much faster.	1A		
			2	

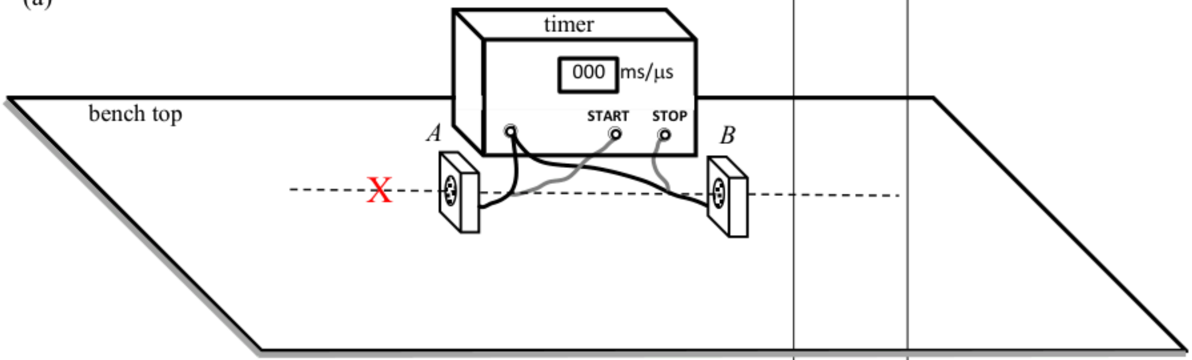
		Poles: 2N	2S/S+N	正	NA
Reason	✓	2	0	1	1
Reason	×/NA	1	0	0	0

NA – Not answered

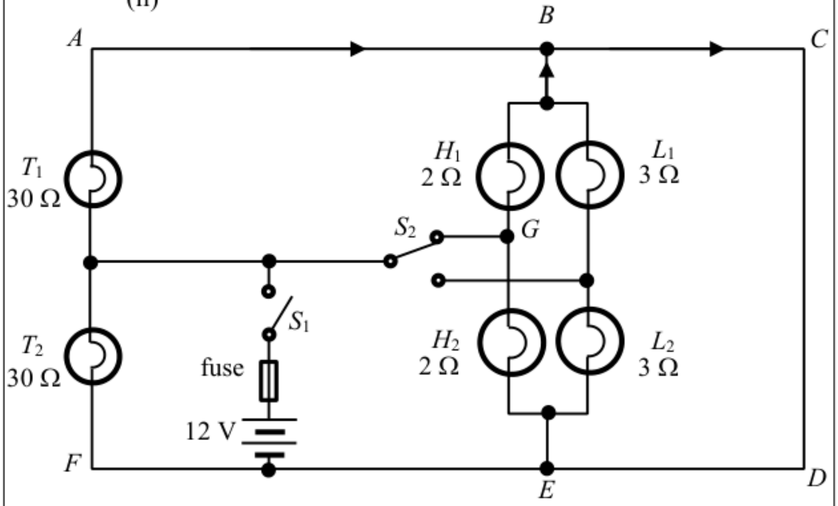
4. (a) $v_B > v_C > v_D > v_A$	1A	[MR: Accept: $v_A < v_D < v_C < v_B$ Accept: B, C, D, A NOT accept: A, D, C, B]
	1	
(b) 	1A 1A	[MR: Accept the arrows drawn at anywhere on the vertical lines passing through the object at positions B & D] a_B correctly marked a_D correctly marked
	2	[MR: 2 marks ：All correct and complete: PE→KE & some KE→PE 1 mark ： (i) PE → KE → PE or (ii) Correctly describe only between 2 points (A→B / B→C / A→C)]
(c) (i) Gravitational potential energy is changed to kinetic energy from A to B, and some kinetic energy is changed back to gravitational potential energy from B to C.	1A 1A	
	2	
(ii) $\frac{1}{2}mv^2 = mgh$ $v^2 = 2(9.81)(2.2 - 1.0)$ $v = 4.852216 \text{ m s}^{-1} \approx 4.85 \text{ m s}^{-1}$	1M 1A	[MR: h must be correct for 1M Method using $v^2 - u^2 = 2as$ etc. is NOT accepted] For $g = 10 \text{ m s}^{-2}$, $v = 4.89898 \text{ m s}^{-1} \approx 4.90 \text{ m s}^{-1}$ Accept 4.85 m s^{-1} to 4.90 m s^{-1}
	2	
(iii) Horizontal speed (at C) = 4.85 × cos 60° $= 2.426108 \text{ m s}^{-1} \approx 2.43 \text{ m s}^{-1}$ Distance = 2.43 × t = 2.55 m t = 1.051066 s ≈ 1.05 s Or Vertical speed (at C) = 4.85 × sin 60° $= 4.2002 \text{ m s}^{-1} \approx 4.20 \text{ m s}^{-1}$ $-1.0 = 4.20 \times t + \frac{1}{2}(-9.81)t^2$ t = 1.051066 s ≈ 1.05 s	1M 1M 1A	e.c.f. from (c)(ii) For $g = 10 \text{ m s}^{-2}$, Horizontal speed $= 2.44949 \text{ m s}^{-1} \approx 2.45 \text{ m s}^{-1}$ Vertical speed ≈ 4.24 m s^{-1} and t = 1.041033 s ≈ 1.04 s Accept 1.04 s to 1.10 s [MR: 1st M for correct expression for initial horizontal /vertical speed 2nd M for correct method (equation) to find t by considering horizontal motion/ vertical motion
	3	

5. (a) (i) 	1A 1A	Forces R(or N) and W(or Mg) correctly indicated and labelled. [MR: 1A for weight with correct label (W/Mg/weight/gravity) 1A for normal reaction with correct label (R/N/normal force/contact force/reaction) 1A only: if both W and R drawn (without/wrong label) Withhold 1A for additional force drawn (more than 2 forces) - W components added with correct labels (ignored) - W components added without correct labels (treated as additional force drawn)
	2	
(ii) $R\cos 30° = W = Mg$ and $R\sin 30° = F$	1M	$M = 10$ kg; $Mg = 98.1$ N [MR: 1M for both equations are correct]
$F = Mg \times \tan 30° = 56.63806 \text{ N} \approx 56.6 \text{ N}$	1A	For $g = 10 \text{ m s}^{-2}$, $57.735027 \text{ N} \approx 57.7 \text{ N}$
$N = R = \dfrac{Mg}{\cos 30°} = 113.27612 \text{ N} \approx 113 \text{ N}$	1A	$115.470054 \text{ N} \approx 115 \text{ N}$
<u>Or</u> $R = W\cos 30° + F\sin 30°$ and $W\sin 30° = F\cos 30°$	(1M)	[MR: 1M for both equations are correct]
	3	
(b) (i) $g\sin\theta = 9.81 \sin 30° = 4.905 \text{ m s}^{-2} \approx 4.91 \text{ m s}^{-2}$	1A	5 m s^{-2} for $g = 10 \text{ m s}^{-2}$,
	1	
(ii) Decrease	1A	
as the component of F perpendicular to the incline no longer acts on the block / incline <u>Or</u> as F is removed, the force pressing the block / incline would decrease (only the weight's component left)	1A	
	2	NA – Not answered / Un - Unchanged

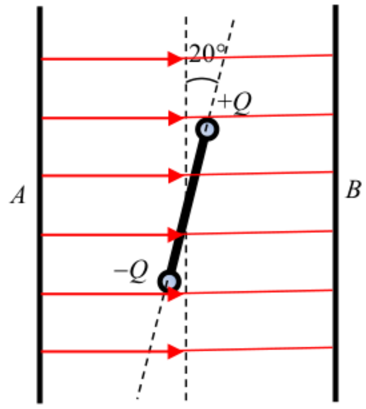
		Answer		
		De	In / Un	NA
Reason	✓	2	0	1
	× / NA	0	0	0
	Not complete	1	0	0

6. (a) 		
Correct location of 'X' (close to axis and on the side of the START microphone).	1A	
Use a metre rule	1A	Accept ruler
to measure the distance / separation D between the two microphones. Record the timer reading, which is the time interval Δt for the sound to travel from one microphone (START) to the other microphone (STOP), i.e. A to B.	1A	Any ONE of the measurements
	3	
(b) (i) Discard 539 µs, $\Delta t = \frac{801+838+821}{3} = 820\ \mu s$ The speed of sound in air $v = \frac{D}{\Delta t}$ $v = \frac{0.280}{820 \times 10^{-6}}$	1M	1M for taking the average value of Δt and quoting the formula $v = \frac{D}{\Delta t}$ to calculate the speed of sound, where D is the separation between the microphones.
$= 341.463415\ \text{m s}^{-1} \approx 341\ \text{m s}^{-1}$	1A	Accept 340 m s^{-1} to 342 m s^{-1}
(ii) Increase separation D between the microphones.	1A	
	3	

7.	(a)	(i)	Angle of incidence at A, $i_A = 90^\circ - 30^\circ = 60^\circ$	1A	
				1	
		(ii)	$n_g \sin c = n_c \sin 90^\circ$	1M	1M for correct equation with angle of refraction in the cladding = 90°
			$\Rightarrow \frac{n_g}{n_c} = \frac{1}{\sin c} \geq \frac{1}{\sin 60^\circ} = 1.1547005 \approx 1.15$	1A	Accept 1.2 or $\frac{2}{\sqrt{3}}$
				2	
		(iii)	Total internal reflection. For light rays from O with $\theta > 30^\circ$, they will fail to undergo total internal reflection.	1A 1A	Correct spelling Accept $\theta \geq 30^\circ$
				2	
	(b)	(i)	Some light / energy (of the narrow light pulse) taking the shortest path (i.e. OD) arrives first while the rest of the energy taking longer paths arrives later. Therefore the pulse is broader and the height of the pulse is lower (smaller intensity).	1A 1A	1A for 'taking different paths' 1A for 'different arrival time' NOT accept: Some of the energy (going at angles larger than 30° from the axis) would be lost due to leakage, thus lower intensity.
				2	
		(ii)	The refractive index of cladding should be increased, as $\frac{n_g}{n_c} = \frac{1}{\sin c}$, by making c / $\sin c$ larger, only those light rays close to the axis / within a smaller θ are totally internal reflected, (such that $\frac{n_g}{n_c}$ getting closer to 1).	1A 1A	0 A for the explanation if the choice for the change of n_c is INCORRECT. Note: More light rays will escape the optical fibre for a larger n_c.
				2	

8. (a) Because L_1 and L_2 (connected in series) are shorted by *BCDE*	1A	Accept p.d. across each low-beam headlight (L_1 or L_2) is 0.
	1	
(b) (i) p.d. across T_2 = 12 V	1A	
	1	
(ii) 	1A 1A	Any two currents correctly marked. All currents correct.
largest current in branch *BC*	1A	
	3	
(c) Power supplied $P = 2 \times \left(\frac{12^2}{30} + \frac{12^2}{2}\right)$	1M	1M for same voltage across each bulb (T_1, T_2, H_1 and H_2)
= 153.6 W = 154 W (3 sig. fig.)	1A	e.c.f. from (b)(i)
$\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6$ (or $\frac{1}{R_{eq}} = \frac{1}{30} + \frac{1}{30} + \frac{1}{2} + \frac{1}{2}$)	1M	1M for all bulbs (T_1, T_2, H_1 and H_2) in parallel
$R_{eq} = 0.9375\ \Omega \approx 0.938\ \Omega \approx 1\ \Omega$ (3 sig. fig.)	1A	
	4	
(d) max. current $I_{max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8$ A (or $I_{max} = \frac{P}{V} = \frac{153.6}{12} = 12.8$ A)	1M	e.c.f. from (c)
Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), it is suitable.	1A	1A for 12.8 A and correct conclusion
	2	

9. (a) **Top view**



9. (a)		Accept: field lines are not on Figure 9.2
	1A	Correct direction (From A to B)
		Accept: field lines not in contact with the plates
	1A	Perpendicular to plates, parallel and evenly spaced
		Accept: 3 or more field lines NOT accept: 2 or more mistakes in the drawing
	2	
(b) (i) $F \times d = (2.0 \times 10^{-5})(0.05 \cos 20°)$	1M	1M $F \times d$ Accept: without consider the component / without component d (0.05, 0.05/2, 2×0.05, 5, 5/2, 2×5) NOT accept: calculation using separation between plates d (0.1, 10)
$= 9.396926 \times 10^{-7}$ N m $\approx 9.40 \times 10^{-7}$ N m	1A	
	2	
(ii) $E = \dfrac{V}{d}$		
$= \dfrac{5.0 \times 10^3}{0.1}$	1M	Accept: $\dfrac{5\times10^3}{0.1}$, $\dfrac{5\times10^3}{1}$, $\dfrac{5}{1}$, $\dfrac{5}{0.1}$
$= 50\ 000$ V m^{-1} or N C^{-1} = 50 kV m^{-1} or kN C^{-1}	1A	
	2	
(iii) $E = \dfrac{F}{Q}$		
$Q = \dfrac{F}{E} = \dfrac{2.0 \times 10^{-5}}{5.0 \times 10^4}$	1M	e.c.f. from (b)(ii)
$= 4.0 \times 10^{-10}$ C	1A	
	2	

10. (a)	proton / ${}^1_1H$ / p / hydrogen nucleus	1A	Accept: proton / ${}^1_1H$ / p / ${}^1_1p$ / H / hydrogen nucleus / hydrogen / hydrogen atom NOT accept: H_2
		1	
(b)	Change in mass $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015)$ $= 0.0012$ u	1M	1M for Δm
	Energy $= 0.0012 \times 931$ $= 1.1172$ (MeV) ≈ 1.12 (MeV) ~~(Or $= 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$~~ ~~$= 1.79388 \times 10^{-13}$ J $\approx 1.79 \times 10^{-19}$ MJ)~~	1A	Accept 1.10 to 1.12 (MeV) Answer should be in MeV, not J. MeV could be omitted [Energy $= 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13}$ J $\approx 1.79 \times 10^{-19}$ MJ ≈ 1.12 (MeV)] ~~Accept (1.79 to 1.80) × 10^{−19} MJ~~
		2	
(c)	By the conservation of momentum, as before the reaction occurs the α particle has momentum, the total momentum of the products (= momentum of the α particle) must also be non-zero,	1A	1st 1A should mention the product(s) has/have momentum 2nd 1A KE/speed of the products must greater than zero Accept: kinetic energy shared by other particles (NOT accept :energy shared.) NOT accept: against repulsion of nucleus / inelastic collisions / KE loss / consider binding energy
	i.e. the total KE of the products must greater than 0, thus the α particle should have a larger KE.	1A	[The kinetic energy of the α particle = energy calculated in (b) + total KE of the products]
		2	